Euclid, Book I, Proposition I.7

On the Same Side of a Given Base There Cannot Exist Two Distinct Vertices Determining Equal Corresponding Sides

Formal Reconstruction — Technical Documentation

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1 Introduction

Proposition I.7 occupies a special position in Book I of Euclid's *Elements*. Unlike the preceding propositions, its principal purpose is not to establish a new geometric construction or a direct property of triangles, but to prove a uniqueness principle.

Euclid shows that, on the same side of a given base, there cannot exist two distinct vertices determining triangles whose corresponding sides are equal. This uniqueness result immediately precedes Proposition I.8, where it is used as a crucial ingredient in the classical proof of the Side–Side–Side congruence theorem.

The Hilbert reconstruction follows a different logical order. The general Side–Side–Side theorem has already been established within the Hilbert interface. Consequently, Proposition I.7 becomes a concise uniqueness argument derived from the developed congruence theory, without reproducing Euclid's original chain of auxiliary arguments.

This proposition therefore provides one of the clearest examples of the difference between the historical organization of Euclid's *Elements* and the logical organization of a modern axiomatic development.

2 The Classical Configuration

Let AB be a given finite straight line and let C and D be two points lying on the same side of the line AB.

Assume that

$$AC \cong AD$$

and

$$BC \cong BD.$$

The proposition asks whether the two vertices C and D can be distinct under these assumptions.

Euclid's objective is to prove that such a configuration is impossible.

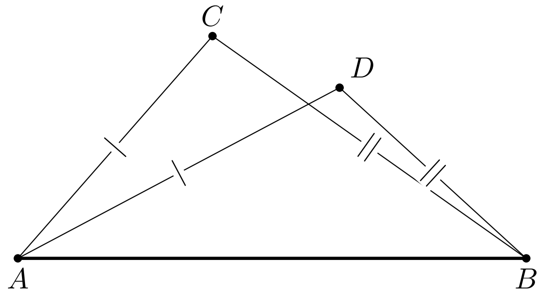

Figure 1: The forbidden configuration of Proposition I.7: distinct points C, D lie on the same side of the base AB, while $AC \cong AD$ and $BC \cong BD$.

2.1 The Contradictory Configuration

Euclid joins the two assumed vertices by the segment CD.

Since

$$AC \cong AD,$$

the triangle ACD is isosceles. By Proposition I.5,

$$\angle ACD \cong \angle CDA.$$

From the geometric position of the points, one of these angles is strictly greater than the corresponding angle formed with the base vertex B.

On the other hand, since

$$BC \cong BD,$$

the triangle BCD is also isosceles. Proposition I.5 therefore gives

$$\angle BCD \cong \angle CDB.$$

The same pair of angles is thus forced to be both equal and unequal, which is impossible. Hence the two assumed vertices cannot be distinct.

3 Statement

Theorem 1 (Euclid I.7)**.** *Let AB be a given base, and let C and D lie on the same side of the line AB.*

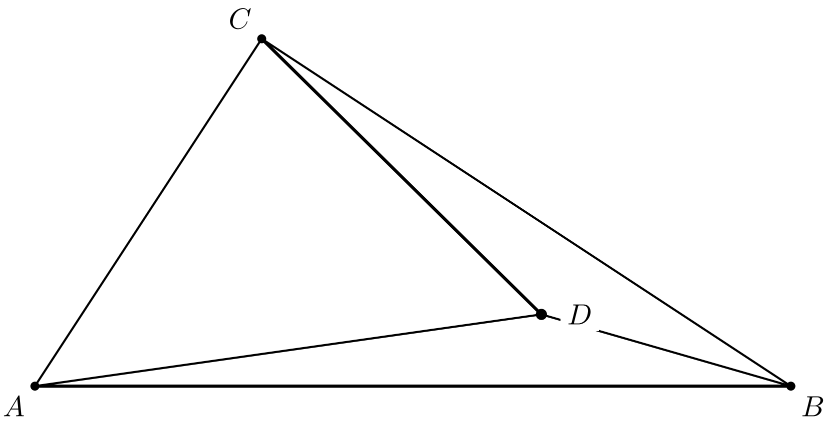

Figure 2: Euclid's representative configuration after joining C to D. The point D is placed deep inside the configuration so that the four side segments and the auxiliary segment CD are visually separated. The equalities $AC \cong AD$ and $BC \cong BD$ are hypotheses of the proposition and are stated in the text rather than marked again on this diagram. This is one positional case, not a diagram of every same-side configuration.

Assume

$$AC \cong AD$$

and

$$BC \cong BD.$$

Then

$$C = D.$$

Equivalently, on a fixed side of a given base there is at most one point having prescribed distances from the two endpoints of that base.

Diagrammatic audit. The two original introductory figures were duplicates and have been merged. One figure now records the actual hypotheses, with the two pairs of equal segments marked; a second records the configuration obtained after joining C and D.

There is an important limitation to the second picture. Wilkins emphasizes that a complete proof of I.7 requires several positional configurations, whereas Euclid explicitly treats only one. In particular, after excluding the cases in which D lies on one of the rays through C, the relevant half-plane is divided into four regions K, L, M, N. The classical diagram below should therefore be read as Euclid's representative case, not as a picture of every possible same-side configuration.

4 Classical Proof

Assume, for contradiction, that C and D are distinct.

Join C to D. Since

$$AC \cong AD,$$

the triangle ACD is isosceles. By Proposition I.5, its base angles are congruent:

$$\angle ACD \cong \angle CDA.$$

Similarly, since

$$BC \cong BD,$$

the triangle BCD is isosceles, and Proposition I.5 gives

$$\angle BCD \cong \angle CDB.$$

The position of the points on the same side of the base forces one of these angles to contain the other as a proper part. Hence an angle would have to be congruent to one of its proper parts, which is impossible.

Therefore the assumption that C and D are distinct leads to a contradiction. Hence

$$C = D.$$

Thus, on a fixed side of a given base, a point is uniquely determined by its distances from the two endpoints of the base.

5 Proof Architecture of the Reconstruction

The formal proof bears little resemblance to Euclid's original argument.

Instead of constructing auxiliary triangles and reasoning by contradiction, the implementation first applies the already established Side–Side–Side theorem of the Hilbert interface. This immediately shows that the two candidate triangles are congruent.

From the resulting congruence, the corresponding angle at the common base vertex is obtained. The uniqueness of angle construction then forces both candidate vertices to lie on the same ray issued from the base vertex.

Finally, since the two points lie on the same ray and are at the same distance from the common endpoint, the uniqueness of segment construction implies that the two points coincide.

```
Euclid
------
same-side distinct C,D
AC ~= AD, BC ~= BD
        |
join CD
        |
I.5 + angle-order argument
```

```
        |
contradiction


Hilbert / Lean
--------------
same-side C,D off base AB
        |
noncollinearity of ABC and ABD
        |
HilbertSSS on ABC and ABD
        |
equal angles at B
        |
hilbert_angle_unique_common_ray
        |
C and D lie on the same ray from B
        |
hilbert_segment_construction_unique
        |
C = D
```

6 Hilbert Reconstruction

The formal proof avoids the positional case analysis highlighted by the classical discussion.

Assume first that $C \neq D$. Since C and D lie on the same side of the line through A, B, both are off that line. The interface theorem `hilbert_not_collinear_of_off_line` therefore gives the noncollinearity of ABC and ABD.

The common side AB, together with

$$AC \cong AD, \qquad BC \cong BD,$$

allows Hilbert SSS to be applied to the triangles ABC and ABD. In particular,

$$\angle ABC \cong \angle ABD.$$

The same-side hypothesis now matters a second time. Angle uniqueness on the common ray at B shows that C and D determine the same ray from B. Finally $BC \cong BD$, together with uniqueness of segment construction on that ray, forces

$$C = D,$$

contradicting the assumption $C \neq D$.

Thus the Hilbert proof replaces Euclid's region-sensitive angle-order argument by two uniqueness principles: uniqueness of the ray determined by an angle on a fixed side, followed by uniqueness of a point at a prescribed distance on that ray.

7 Lean Formalization

The complete implementation is available in the repository. Three short excerpts expose the structure of the formal argument.

First SSS is applied to the two triangles sharing the base AB:

```
have hSSS :=
  HilbertSSS
    Geo
    A B C
    A B D
    hABC
    hABAB
    hBCBD
    hACAD
```

The angle at B is then fed into the same-ray uniqueness theorem:

```
rcases
    hilbert_angle_unique_common_ray
      Geo
      A B C D
      base
      hAB
      hAbase
      hBbase
      hCbase
      hSame
      hAngleB with
  <X, hRayXC, hRayXD>
```

Finally, the equal distances from B and the common ray identify the two points:

```
exact
  hilbert_segment_construction_unique
    Geo
    B C
    B X
    C D
    hRayXC
    hRayXD
    (hilbert_congruent_reflexive Geo B C)
    hBD_BC
```

The remaining code establishes the off-line and noncollinearity conditions required by these interface theorems. Those bookkeeping steps are important for verification but do not add a new geometric idea to the proof.

8 Relationship with Earlier Propositions

8.1 Proposition I.5

The classical proof of I.7 creates isosceles triangles from the assumed equal distances to the endpoints of the common base. Proposition I.5 then converts those segment congruences into base-angle congruences.

This is the first essential step in the contradiction: metric symmetry at the level of sides becomes angular symmetry.

8.2 Proposition I.6

Proposition I.6 is the converse of the isosceles-triangle theorem. It allows angle congruence to be converted back into segment congruence when the contradiction argument reaches the appropriate auxiliary triangle.

Thus I.5 and I.6 together provide the two-way bridge between side congruence and angle congruence used in the uniqueness argument.

8.3 Transition to Proposition I.8

The importance of I.7 becomes fully visible in Proposition I.8.

Euclid's SSS proof superposes one triangle on another. After two vertices and the common base have been matched, I.7 excludes the possibility that the third vertex could occupy a different position on the same side of the base while preserving both distances to the base endpoints.

Thus I.7 supplies the uniqueness principle hidden inside Euclid's superposition proof of SSS.

9 Hilbert and Book Zero Components Used

9.1 Same-Side Geometry

The statement of I.7 is fundamentally a same-side uniqueness theorem. The two candidate vertices are assumed to lie on the same side of the common base. This positional information is explicit in the Hilbert reconstruction rather than inferred from the drawing.

9.2 Segment Congruence

The hypotheses prescribe equal distances from each candidate vertex to the two endpoints of the base. The formal proof repeatedly transports and composes these congruences in order to create the isosceles configurations used in the contradiction.

9.3 Isosceles Angle Theorems

The established Hilbert isosceles infrastructure converts the equal-side hypotheses into angle congruences. These are the formal counterparts of Euclid's applications of I.5.

9.4 Order and Collinearity

The contradictory configuration depends on the relative positions of the candidate vertices and the base. Betweenness, collinearity, and same-side information replace the diagrammatic assumptions in the classical proof.

10 Formalization Notes

Proposition I.7 illustrates a fundamental difference between the historical development of Euclid's geometry and its modern axiomatic reconstruction.

In the *Elements*, Proposition I.7 serves as an essential lemma for the proof of the Side–Side–Side congruence theorem in Proposition I.8. Euclid therefore establishes the uniqueness result before proving the general congruence theorem.

The Hilbert reconstruction follows the opposite logical direction. Once the general Side–Side–Side theorem has been derived from the Hilbert axioms, Proposition I.7 becomes an immediate uniqueness corollary obtained by combining triangle congruence with the uniqueness of angle and segment constructions.

This proposition therefore demonstrates that the historical order of the *Elements* need not coincide with the logical organization of a modern formal theory. As the interface develops, results that were once foundational become concise consequences of more general synthetic theorems.

11 Dependency Summary

The geometric content of Proposition I.7 can be summarized as

$$\begin{gathered} C, D \text{ on the same side of } AB \\ AC \cong AD, \qquad BC \cong BD \\ \downarrow \\ \text{isosceles configurations} \\ \downarrow \\ \text{I.5 and angle-congruence transport} \\ \downarrow \\ \text{contradictory angle configuration} \\ \downarrow \\ C = D. \end{gathered}$$

Equivalently, on a prescribed side of a base, the two distances to the base endpoints determine the vertex uniquely.

The formal reconstruction uses only established incidence, order, and congruence infrastructure. No coordinate representation or numerical distance is introduced.

12 Conclusion

Proposition I.7 is best understood as a rigidity or uniqueness theorem.

Its classical formulation is somewhat indirect: two different points on the same side of a base cannot have the same prescribed distances to both endpoints of that base. The formal reconstruction exposes the structural content more clearly: the intersection of two distance conditions is unique once the side of the base has been fixed.

The proof depends on the isosceles theory developed in I.5 and I.6 and turns segment congruence into a contradiction among angles and point positions.

Its main importance is prospective. Proposition I.8 uses exactly this uniqueness principle to justify the final step of the SSS superposition argument. I.7 therefore forms the bridge between the elementary isosceles results and the general triangle-rigidity theorem of I.8.